

Space Complexity

Space is a computation resource. Unlike time it can be reused.

Synopsis

1. Space Bounded Computation
2. Logspace Reduction
3. PSPACE Completeness
4. Savitch Theorem
5. NL Completeness
6. Immerman-Szelepcsényi Theorem

Space Bounded Computation

Space Bounded Computation

Let $S : \mathbf{N} \rightarrow \mathbf{N}$ and $L \subseteq \{0, 1\}^*$.

We say that $L \in \mathbf{SPACE}(S(n))$ if there is some c and some TM deciding L that never uses more than $cS(n)$ nonblank worktape locations on inputs of length n .

Space Constructible Function

Suppose $S : \mathbf{N} \rightarrow \mathbf{N}$ and $S(n) \geq \log(n)$.

1. S is **space constructible** if there is a Turing Machine that computes the function $1^n \mapsto \lfloor S(n) \rfloor$ in $O(S(n))$ space.
2. S is **space constructible** if there is a Turing Machine that upon receiving 1^n uses exactly $S(n)$ -space.

The second definition is slightly less general than the first.

Space Bounded Computation, the Nondeterministic Case

$L \in \mathbf{NSPACE}(S(n))$ if there is some c and some NDTM deciding L that never uses more than $cS(n)$ nonblank worktape locations on inputs of length n , **regardless of** its nondeterministic choices.

For space constructible function $S(n)$ we could allow a machine in $\mathbf{NSPACE}(S(n))$ to diverge and to use more than $cS(n)$ space in unsuccessful computation paths.

Configuration

A configuration of a running TM M with input x consists of the following:

- ▶ the state;
- ▶ the content of the **work** tape; [In the study of space complexity one may always assume that there is one work tape.]
- ▶ the head positions.

We write C_{start} for the unique initial configuration.

We assume that there is a single accepting configuration C_{accept} .

Configuration Graph

A configuration graph $G_{M,x}$ of M with input x is a directed graph:

- ▶ the nodes are configurations;
- ▶ the arrows are one-step computations.

“ M accepts x ” iff “there is a path in $G_{M,x}$ from C_{start} to C_{accept} ”.

Reachability Predicate for Configuration Graph

Suppose $\mathbb{M}$ is an $S(n)$ -space TM.

- ▶ A vertex of $G_{\mathbb{M},x}$ is described using $O(S(|x|))$ bits.
- ▶ Therefore $G_{\mathbb{M},x}$ has at most $2^{O(S(|x|))}$ nodes.

There is an $O(S(n))$ -size CNF $\varphi_{\mathbb{M},x}$ such that for every two configurations C and C' , $\varphi_{\mathbb{M},x}(C, C') = 1$ iff $C \rightarrow C'$ is an edge in $G_{\mathbb{M},x}$.

- ▶ $\varphi_{\mathbb{M},x}(C, C')$ can be checked by essentially comparing C and C' bit by bit. It can be accomplished in both
 - ▶ $O(S(n))$ time, and
 - ▶ $O(\log S(n))$ space.

Space vs. Time

Theorem. Suppose $S(n) : \mathbf{N} \rightarrow \mathbf{N}$ is space constructible. Then

$$\mathbf{TIME}(S(n)) \subseteq \mathbf{SPACE}(S(n)) \subseteq \mathbf{NSPACE}(S(n)) \subseteq \mathbf{TIME}(2^{O(S(n))}).$$

An algorithm proving that $\mathbf{NSPACE}(S(n)) \subseteq \mathbf{TIME}(2^{O(S(n))})$ starts by constructing $G_{\mathbb{M},x}$ in $2^{O(S(n))}$ time, and then applies the breadth first search algorithm to $G_{\mathbb{M},x}$.

Space vs. Time

Theorem. For all space constructible $S(n)$, $\mathbf{TIME}(S(n)) \subseteq \mathbf{SPACE}(S(n)/\log S(n))$.



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1. Hopcroft, Paul and Valiant. On Time versus Space and Related Problems. FOCS, 1975.

Space Complexity Class

$$\begin{aligned}\mathbf{PSPACE} &\stackrel{\text{def}}{=} \bigcup_{c>0} \mathbf{SPACE}(n^c), \\ \mathbf{NPSPACE} &\stackrel{\text{def}}{=} \bigcup_{c>0} \mathbf{NSPACE}(n^c), \\ \mathbf{L} &\stackrel{\text{def}}{=} \mathbf{SPACE}(\log(n)), \\ \mathbf{NL} &\stackrel{\text{def}}{=} \mathbf{NSPACE}(\log(n)).\end{aligned}$$

Games are Harder than Puzzles

$\text{NP} \subseteq \text{PSPACE}.$

Example

The following problems are in **L**:

EVEN $\stackrel{\text{def}}{=} \{x \mid x \text{ has an even number of } 1\text{'s}\},$

PLUS $\stackrel{\text{def}}{=} \{(\ulcorner m \urcorner, \ulcorner n \urcorner, \ulcorner m + n \urcorner) \mid m, n \in \mathbf{N}\},$

MULP $\stackrel{\text{def}}{=} \{(\ulcorner m \urcorner, \ulcorner n \urcorner, \ulcorner m \times n \urcorner) \mid m, n \in \mathbf{N}\}.$

PATH is in NL

PATH = $\{\langle G, s, t \rangle \mid \text{there is a path from } s \text{ to } t \text{ in the digraph } G\}$.

Theorem. $\text{PATH} \in \text{NL}$.

Proof.

Both a node and a counter can be stored in logspace.



Universal Turing Machine without Space Overhead

Theorem. There is a universal TM that operates without space overhead for input TM's with space complexity $\geq \log(n)$.

A universal TM can simulate M_α by recording all the non-blank tape content of M_α in its single work tape. A counter is used to store the location of the reader at x .

Some additional space, whose size depends only on M_α , is needed for bookkeeping.

Space Hierarchy Theorem

Theorem. If f, g are space constructible such that $f(n) = o(g(n))$, then

$$\mathbf{SPACE}(f(n)) \subsetneq \mathbf{SPACE}(g(n)).$$

We design $\mathbb{V}$ by modifying the universal machine so that

- ▶ $\mathbb{V}(x)$ simulates $\mathbb{M}_x(x)$, and
- ▶ it stops when it is required to use more than $g(n)$ space, and
- ▶ it negates the result after it completes simulation.

If $\mathbb{V}$ was executed in $f(n)$ space, then $\mathbb{V} = \mathbb{M}_\alpha$ for some large enough α so that $\mathbb{V}$ can complete the simulation of $\mathbb{M}_\alpha$ on α .

But then $\overline{\mathbb{M}_\alpha(\alpha)} = \mathbb{V}(\alpha) = \mathbb{M}_\alpha(\alpha)$.

1. J. Hartmanis and R. Stearns. On the Computational Complexity of Algorithms. Transactions of AMS, 117:285-306, 1965.

Logspace Reduction

Logspace Reduction

A function $f : \{0, 1\}^* \rightarrow \{0, 1\}^*$ is **implicitly logspace computable** if the following hold:

1. $\exists c. \forall x. |f(x)| \leq c|x|^c$,
2. $\{\langle x, i \rangle \mid i \leq |f(x)|\} \in \mathbf{L}$ and
3. $\{\langle x, i \rangle \mid f(x)_i = 1\} \in \mathbf{L}$.

Problem B is **logspace reducible** to problem C , written $B \leq_L C$, if there is an implicitly logspace computable f such that $x \in B$ iff $f(x) \in C$.

- ▶ Logspace reductions are Karp reductions. [The converse implication is unknown.]
- ▶ All known NP-completeness results can be established using logspace reduction.

Transitivity of Logspace Reduction

Lemma. If $B \leq_L C$ and $C \leq_L D$ then $B \leq_L D$.

Let M_f, M_g be logspace machines that compute $x, i \mapsto f(x)_i$ respectively $y, j \mapsto g(y)_j$. We construct a machine that, given input x, j with $j \leq |g(f(x))|$, outputs $g(f(x))_j$.

- ▶ The machine operates as if $f(x)$ were stored on a **virtual** tape.
 - ▶ It stores the address i of the current cell of the virtual tape.
 - ▶ It uses $O(\log |f(x)|) = O(\log |x|)$ space to calculate $g(f(x))_j$.

Logspace Computability

A function $f : \{0, 1\}^* \rightarrow \{0, 1\}^*$ is **logspace computable** if it can be computed by a TM that has a **write-once output tape** using $O(\log n)$ work tape space.

Lemma. Implicitly logspace computability = logspace computability.

PSPACE Completeness

Space Completeness

A language L' is **PSPACE**-hard if $L \leq_L L'$ for every $L \in \mathbf{PSPACE}$.

If in addition $L' \in \mathbf{PSPACE}$ then L' is **PSPACE**-complete.

Quantified Boolean Formula

A **quantified Boolean formula** (QBF) is a formula of the form

$$Q_1x_1 Q_2x_2 \dots Q_nx_n \cdot \varphi(x_1, \dots, x_n)$$

where each Q_i is one of the two quantifiers $\forall, \exists$, $x_1, \dots, x_n$ range over $\{0, 1\}$, and φ is a quantifier free Boolean formula containing no free variables other than $x_1, \dots, x_n$.

Let **TQBF** be the set of **true** QBFs.

Quantified Boolean Formula

Suppose φ is a CNF.

- ▶ $\varphi \in \text{SAT}$ if and only if $\exists \tilde{x}. \varphi \in \text{TQBF}$.
- ▶ $\varphi \in \text{TAUTOLOGY}$ if and only if $\forall \tilde{x}. \varphi \in \text{TQBF}$.

Stockmeyer-Meyer Theorem. TQBF is **PSPACE**-complete.



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1. Larry Stockmeyer, Albert Meyer. Word Problems Requiring Exponential Time. STOC, 1973.

Proof of Stockmeyer-Meyer Theorem

TQBF $\in$ **PSPACE**.

Suppose $\psi = Q_1x_1 Q_2x_2 \dots Q_nx_n.\varphi(x_1, \dots, x_n)$.

- ▶ A counter of length n can be identified to an assignment.
- ▶ Apply the **depth first** tree traversal.

We actually have a linear space algorithm.

Proof of Stockmeyer-Meyer Theorem

Let $\mathbb{M}$ be a TM that decides L in polynomial space, say $S(n)$ space.

We reduce $x \in \{0, 1\}^*$ to a QBF φ_x of size $O(S(|x|)^2)$ in **logspace** such that $\mathbb{M}(x) = 1$ iff φ_x is true.

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1. Construct in logspace ψ_0 such that $\psi_0(C, C')$ is true if and only if $C \rightarrow C'$.
 2. Let $\psi_i(C, C')$ be true if and only if there exists a path of **length** $\leq 2^i$ from C to C' . It can be defined by the following formula, which is computable in logspace.

$$\exists C'' \forall D^1 \forall D^2. ((D^1 = C \wedge D^2 = C'') \vee (D^1 = C'' \wedge D^2 = C')) \Rightarrow \psi_{i-1}(D^1, D^2).$$

3. Now $|\psi_i| = |\psi_{i-1}| + O(S(|x|))$. Hence $|\varphi_x| = |\psi_{S(|x|)}| = O(S(|x|)^2)$.

QBF Game

Two players make alternating moves on the board

$$\exists x_1 \forall x_2 \exists x_3 \forall x_4 \cdots \exists x_{2n-1} \forall x_{2n} \cdot \varphi(x_1, \dots, x_{2n}).$$

Player I moves first. It has a **winning strategy** if φ is true after Player II's last move, no matter how Player II plays.

- ▶ Deciding if Player I has a winning strategy for QBF game is **PSPACE**-complete.

1. Christos Papadimitriou. Games Against Nature. FOCS, 1983.

Savitch Theorem



Savitch Theorem. If S is space constructible then $\mathbf{NSPACE}(S(n)) \subseteq \mathbf{SPACE}(S(n)^2)$.

Suppose $\mathbb{N}$ is an NDTM that decides L in $S(n)$ space.

Given $x \in \{0, 1\}^*$, a divide-and-conquer depth first algorithm can be designed that searches for a path from C_{start} to C_{accept} in $G_{\mathbb{N}, x}$.

The depth of the recursive calls is $S(|x|)$.

1. W. Savitch. Relationships between Nondeterministic and Deterministic Tape Complexities. JCSS, 177-192, 1970.

$$\text{PSPACE} = \text{NPSPACE}$$

NL Completeness

NL-Completeness

C is **NL**-complete if it is in **NL** and $B \leq_L C$ for every $B \in \mathbf{NL}$.

NL-Completeness

Theorem. PATH is **NL**-complete.

Suppose a nondeterministic TM $\mathbb{N}$ decides L in $O(\log(n))$ space. A logspace reduction from L to PATH is defined by the following reduction:

$$x \mapsto \langle G_{\mathbb{N},x}, C_{\text{start}}, C_{\text{accept}} \rangle.$$

The graph $G_{\mathbb{N},x}$ is represented by an **adjacent matrix**, every bit of it can be calculated in $O(|C|) = O(\log |x|)$ space.

We may assume that all space bounded TM's terminate by making use of counters. It follows that PATH remains **NL**-complete if only **acyclic** graphs are admitted.

NL is nothing but PATH.

Immerman-Szelepcsényi Theorem

Savitch Theorem implies **coNPSPACE = NPSPACE**.

But **coNL = NL** is a different story.

Immerman-Szelepcsényi Theorem. $\overline{\text{PATH}} \in \text{NL}$.



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1. R. Szelepcsényi. The Method of Forcing for Nondeterministic Automata. Bulletin of EATCS, 1987.
 2. N. Immerman. Nondeterministic Space is Closed under Complementation. SIAM Journal Computing, 1988.

Proof of Immerman-Szelepcsényi Theorem

Design a logspace NDTM $\mathbb{N}$ such that for vertices s, t of a graph G with n -vertices, $\mathbb{N}(\langle G, s, t \rangle) = 1$ iff there is **red** path from s to t .

For $i \in [n-1]$,

- ▶ let C_i be the set of nodes reachable from s in i -steps, and [By definition $C_1 \subseteq C_2 \subseteq \dots \subseteq C_{n-1}$.]
- ▶ let $c_i = |C_i|$. Notice that c_1 can be stored in logspace.

We **can** store a fixed number of c_i 's, but we **cannot** store any of C_i 's.

Set $c_{i+1} = 0$. For each vertex $v \neq s$, $\mathbb{N}$ **guesses** C_i and increments c_{i+1} if either $v \in C_i$ or $u \rightarrow v$ for some $u \in C_i$.

- ▶ For each u in C_i check that s reaches to u in i steps. This is PATH.
 - ▶ A counter is maintained to ensure that $|C_i| = c_i$.
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After c_{n-1} has been calculated, guess C_{n-1} and accept if $t \notin C_{n-1}$.

Immerman-Szelepcsényi Theorem

Corollary. For every space constructible $S(n) \geq \log(n)$, one has

$$\mathbf{coNSPACE}(S(n)) = \mathbf{NSPACE}(S(n)).$$

Theorem. 2SAT is **NL**-complete.

Given $\langle G, s, t \rangle$, where G is **acyclic**, we translate an edge $x \rightarrow y$ to the clause $\bar{x} \vee y$. We also add clauses s and $\bar{t}$. This is a logspace reduction from acyclic $\overline{\text{PATH}}$ to 2SAT.

By Hierarchy Theorems some of the following inclusions are strict.

$$\mathbf{L} \subseteq \mathbf{NL} \subseteq \mathbf{P} \subseteq \mathbf{NP} \subseteq \mathbf{PSPACE} \subseteq \mathbf{EXP}.$$

Yet we don't know which is strict.

- ▶ It is widely believed that $\mathbf{NL} \subsetneq \mathbf{P}$.
- ▶ “ $\mathbf{L} \stackrel{?}{=} \mathbf{NL}$ ” is a major open problem in the structural theory.